URL Safety Checker

Python Source Code

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"""
URL Safety Checker
Educational cybersecurity mini-project.

This tool checks URL structure and common warning signs.
It does NOT guarantee that a URL is safe.
"""

from urllib.parse import urlparse
import re

SUSPICIOUS_WORDS = {
    "login", "verify", "update", "secure", "account",
    "password", "banking", "confirm", "free", "prize"
}

def check_url(url):
    result = {
        "url": url,
        "valid_format": False,
        "uses_https": False,
        "hostname": "",
        "warnings": []
    }

    url = url.strip()
    if not url:
        result["warnings"].append("No URL was entered.")
        return result

    # Add a scheme for parsing when the user enters only a domain.
    candidate = url if "://" in url else "https://" + url
    parsed = urlparse(candidate)
    hostname = parsed.hostname or ""

    if hostname and "." in hostname and " " not in hostname:
        result["valid_format"] = True
        result["hostname"] = hostname
    else:
        result["warnings"].append("The URL format looks invalid.")

    result["uses_https"] = parsed.scheme.lower() == "https"
    if not result["uses_https"]:
        result["warnings"].append("The URL does not use HTTPS.")

    if "@" in parsed.netloc:
        result["warnings"].append(
            "The address contains '@', which can be used to disguise a destination."
        )

    if re.match(r"^\d{1,3}(\.\d{1,3}){3}$", hostname):
        result["warnings"].append(
            "The hostname is an IP address rather than a regular domain."
        )
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    if len(url) > 100:
        result["warnings"].append("The URL is unusually long.")

    if hostname.count("-") >= 3:
        result["warnings"].append("The domain contains many hyphens.")

    lower_url = url.lower()
    found_words = sorted(word for word in SUSPICIOUS_WORDS if word in lower_url)
    if found_words:
        result["warnings"].append(
            "Contains words often seen in misleading links: " + ", ".join(found_words)
        )

    if not result["warnings"]:
        result["warnings"].append(
            "No basic warning signs were detected. This does not prove the URL is safe."
        )

    return result

def main():
    print("=" * 45)
    print("         URL SAFETY CHECKER")
    print("=" * 45)
    url = input("Enter a URL to check: ")
    report = check_url(url)

    print("\n--- CHECK REPORT ---")
    print("URL:", report["url"])
    print("Format appears valid:", "Yes" if report["valid_format"] else "No")
    print("Uses HTTPS:", "Yes" if report["uses_https"] else "No")
    print("Hostname:", report["hostname"] or "Not detected")
    print("\nWarnings:")
    for number, warning in enumerate(report["warnings"], start=1):
        print(f"{number}. {warning}")

    print("\nReminder: this is a basic educational checker, not a malware scanner.")

if __name__ == "__main__":
    main()
```